

UNIVERSITY of WASHINGTON

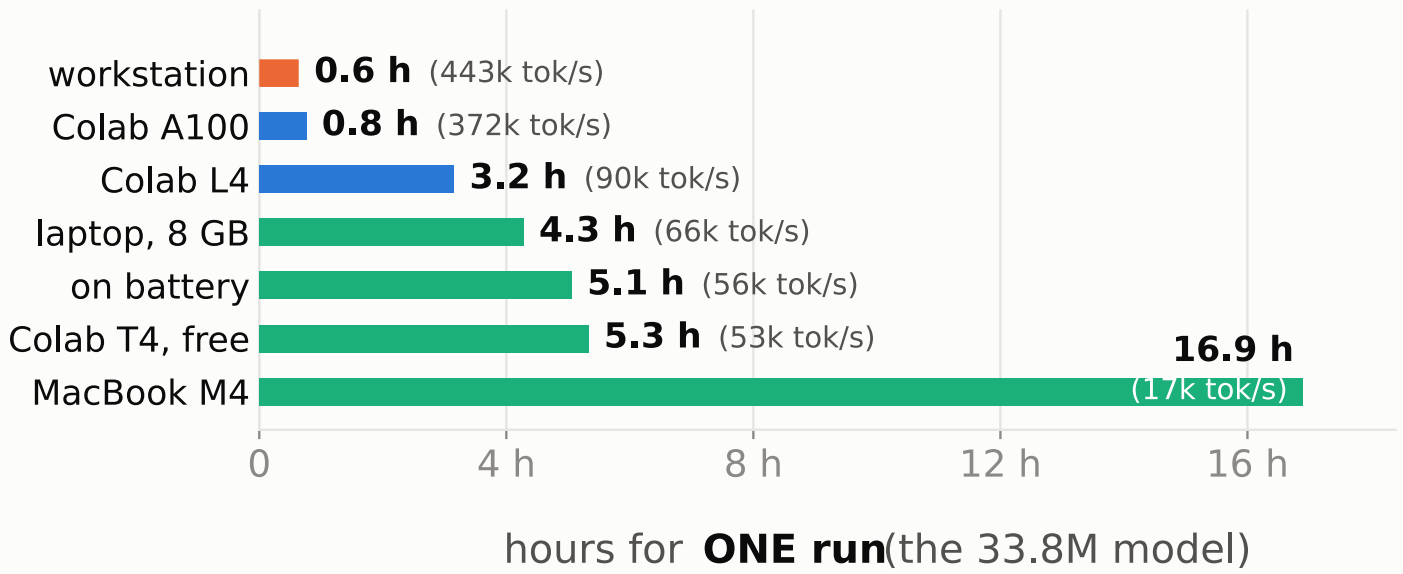
CSED 505: BUILDING A MODEL
FACTORY



501 the statistics · 502 the mechanics · 503 the language stack · 504 scale — All four end when a model finishes training. None covers needing a hundred models and having to believe the differences between them.

Jeffrey Stall · Patrick Kwok · Leon Wan — CSED 504. This board is the machinery; the upper board is the experiment it served.

You do not need the workstation

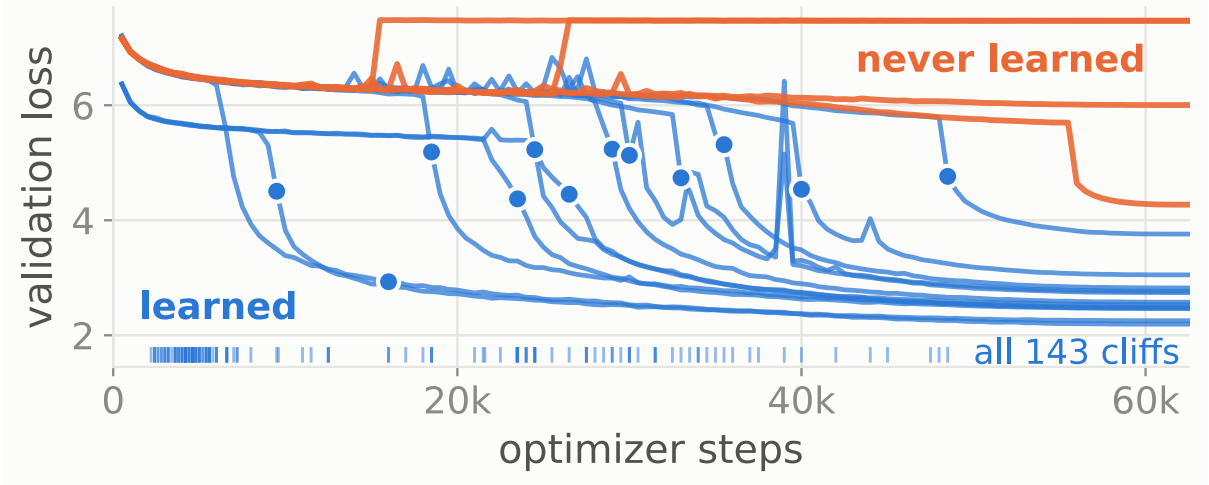


An 8 GB laptop GPU beats the free tier, even unplugged. The two paid Colab tiers bill the same — \$111 for 7 days against \$112 for 30 — so the faster tier buys time, not money. Electricity for all 148 GPU-hours was \$7; the workstation that ran them, \$24,000.

Budget in tokens, not epochs

Epochs were right in 502–504: the dataset was fixed. Here the dataset *is* the experiment — "40 epochs each" gives one arm 256× the compute of another. So fix the total number of tokens, and let epochs fall out. Every run is 1.024B tokens: the 4M corpus is seen 256 times, the 1,024M once. Tokens + tok/s = the wall clock. 1.024B tokens ÷ 443k tok/s ≈ 38 min — the top bar of the chart above. A truncated run is a different experiment. The schedule anneals to zero at the *planned* end — a stopped run is caught mid-schedule, not half as good. Our own records carried the bug. The epoch field reads 125; the truth is 16 passes — the name outlived the meaning.

You cannot judge a run early



Yoruba, English and ten more, all on one axis. Every run that learns sits flat and then falls off a cliff — and no two fall at the same step: earliest 2,200, latest 48,500. So "still flat at step k" is evidence of nothing. 35 of 197 never learned.

A hundred runs is a filing problem first

A study is declared as a grid — languages × rates × seeds — never typed run by run. The queue orders it longest-budget-first. Every run writes a record naming everything that changes the answer — settings, data fingerprint — after two runs differing only in vocabulary silently overwrote each other. "What have we tried?" is a query: 1,089 runs later, no collisions, no archaeology.

A factory that makes one thing is not a factory

English is the ruler. *Why*: the one language where data was never the constraint. *Learned*: more data stops helping past 64M tokens — which Yoruba, having no more, could never show. French, Indonesian, Mandarin are the control. *Why*: XLM-R knows all three well. *Learned*: they cost 1.04, 1.01, 0.95 — the penalty tracks coverage, not "African". Twelve African languages are the gradient. *Why*: an anecdote needs a slope behind it. *Learned*: the split falls along coverage — with Wolof, uncovered yet cheap at 1.31, the exception an anecdote hides. Five languages got the whole sweep. *Why*: "one call per language" is only true if the settings transfer. *Learned*: they do not — the rate four languages want destroys Igbo. None of us speaks any of them. So the spread is the evidence, not any one score. Nine functions, one call per language — twelve pretrained in 48 minutes.

When memory is a wall, and when it is only a speed limit

What has to fit is the model — 12–16 bytes a parameter. Our 98M model: 1.6 GB. A 7B model: 84–112 GB — no consumer card has it. Everything above the floor is batch, and batch is optional. Accumulation runs a big step as smaller passes — the same update, within 2% of the cost. So an 8 GB laptop trains the step a 96 GB card trains. A 10 GB run folds into 6.1 GB at 28,027 tok/s. Nobody on the chart beside this needed 80 GB. 96, 80, 24, 16, 8 GB — all ran the same step. The A100 sells speed, not room. And Windows does not warn you. An oversized batch spills to system RAM: 28,027 → 5,075 tok/s, no error.

What we made faster

2.07× on the four-cell benchmark — only 1.32× of it efficiency; the rest is a second card nobody had used. A third of the twenty were rejected, and a list of only the wins would be marketing rather than a log. Every row had to make the same experiment faster.

WHAT CHANGED	MEASURED	KEPT?
THE DATA PATH		
1 · dataset on the card, no loader	13.8k → 18.5k img/s	yes
2 · images as uint8, cast per batch	4× less memory, free cast	yes
3 · augment on the GPU	no PIL, no host copy	yes
4 · tokenize once, up front	53 s vs 85 min — 96×	yes
5 · token dtype from the vocabulary	16k → 2 bytes a token	yes
PRECISION AND KERNELS		
6 · bf16 autocast, not fp16	1.34× on sm_89, no loss-scale	yes
7 · channels_last for CNNs, bf16 only	fp16 + it is 3.5× slower	guarded
8 · channels_last for ViTs	a no-op on a transformer	no
9 · TF32 for matmul and cuDNN	free on Ampere and later	yes
10 · cudnn.benchmark = True	worth it when shapes are fixed	yes

WHAT CHANGED	MEASURED	KEPT?
11 · fused SGD and AdamW	one kernel, not a chain	yes
12 · torch.compile	no Triton on Windows	can't
13 · attention backend	already sdpa	no change
HOST-SYNC DISCIPLINE		
14 · drop the per-step .item(), CIFAR	~7%	yes
15 · the same at MLM scale	2% — a live loss is worth it	no
16 · zero_grad(set_to_none=True)	skips a full zero-fill	yes
BATCHING AND SCHEDULING		
17 · batch 64 → 128	1.31× — 2048 is slower	yes
18 · checkpointing for speed	peak is 3.3 GB of 96	no
19 · use the second card	91% / 93% busy at 300 W	yes
20 · longest-budget-first queue	2.55× projected, 2.07 seen	yes

The setting four languages want destroys the fifth

Yoruba	✗	+0.22	★	+0.02	+0.19	🌙
Swahili	✗	+0.55	+0.09	★	+0.06	+0.16
Nyanja	✗	+0.70	+0.12	★	+0.05	+0.64
Hausa	✗	+0.38	+0.07	★	+0.02	+0.05
Igbo	🌙	+0.07	★	✗	✗	✗
	1.5	3	5	7	8.5	10
	peak learning rate (×10 ⁻⁴)					

Five languages, six learning rates, sixty runs, each cell scored against that language's own best: ★ its winner, ✗ far behind it, 🌙 one seed of two. Four land at their best in the outlined column. Igbo collapses there — the same setting, a wasted night.

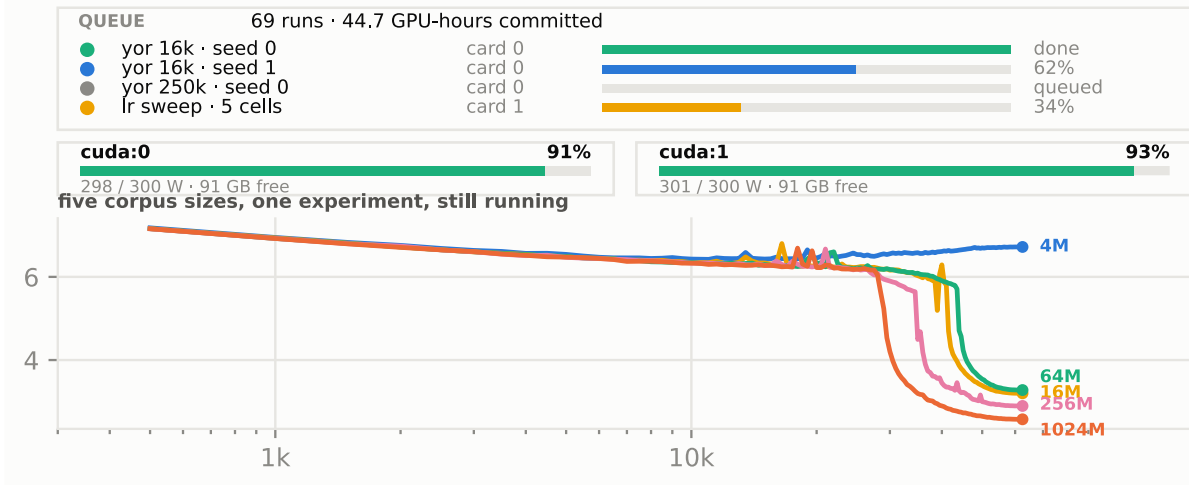
A result has to beat your own noise

Run the same recipe twice and it lands in two different places. That is the seed, not the change you made. So we measured our own noise first and made every claim clear it by 2.27× — at three runs a side, no test can call anything significant. The spread is a property of each cell — re-measured, never a constant we reused. Two of our own claims did not clear it. We retired them.

The factory, in numbers

- 197 pretraining runs, every one from scratch
- 892 fine-tuning runs on top of them, across two tasks
- 17 languages measured, 12 of them pretrained from scratch
- 148 GPU-hours, two cards, 71 kWh at the wall
- 62,500 steps a run — 1.024B tokens of updates, the unit that makes two machines comparable
- 0 identity collisions, because a vocabulary carries a hash

One screen, both cards, a rush order at 9 p.m.



The chart is five English corpus sizes separating while they run. On 9 August 44.7 GPU-hours sat across 69 runs and Patrick needed a sweep: a fleet pins to one card, so it took card 1 while the queue kept card 0. Twenty minutes to running, on a measured promise.

Ethics: what we do not claim, and what is next

Nobody on this team reads Yoruba — every judgement here is a benchmark number. The corpus is scraped web text nobody consented to. The audit gate prints 9 claims: 6 supported, 2 not, 1 underpowered. Next: a wall meter, a corpus audit, the evaluation we could not do.

Sources

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